

Full Length Article

Lightweight attention-aware fusion network based on state-space model for V-D-T salient object detection ^{*}

Anzhi Wang ^{a,*}, Jintao Wu ^b, Yun Liu ^c

^a School of Bigdata and computer science, Guizhou Normal University, Guiyang, 550025, Guiyang, China

^b Frontiers Science Center for Smart High-speed Railway System, Beijing Jiaotong University, Beijing, 100044, Beijing, China

^c College of Artificial Intelligence, Southwest University, Chongqing, 400715, Chongqing, China

ARTICLE INFO

Keywords:

Salient object detection
Visible-depth-thermal
Lightweight network
State spatial model
Attention

ABSTRACT

With the popularization of various sensors, multimodal Salient Object Detection (SOD) methods including Visual-Depth-Thermal Salient Object Detection (V-D-T SOD) have made remarkable developments. However, most existing V-D-T SOD methods usually achieve accurate detection with expensive computational cost, which limits their development at the edge application. To address this issue, we develop a Lightweight Attention-Aware Fusion Network (LAANet) based on the State Space Model (SSM). Specifically, inspired by the cross-attention mechanism, the Cross Mamba Fusion Module (CMFM) is proposed to realize attention perception among shallow high-resolution multimodal features by exploiting SSM with linear complexity. For low-resolution deep features, Attention Perception Fusion Module (APFM) is proposed based on the self-attention mechanism to mine semantic cues and complement the potential information lost in SSM amnesic memory. In addition, to response the problem of excessive decoding parameters and efficiently achieve feature reconstruction, we design a lightweight decoder (LDB). It is consist of simple dilated convolutions and linear operations with only 0.37M parameter. Extensive experiments on the VDT2048 dataset show that our method achieves performance close to that of SOTA, while having smaller number of parameters (6.78M), lower model complexity (5.23G), and faster inference speed (23.7FPS when the input size is 384*384). The code is available at <https://github.com/GZNU-WJT/LAANet>.

1. Introduction

As a fundamental task in computer vision, Salient Object Detection (SOD) aims to endow computers with the ability to perceive the most notable objects in the scenario. SOD can be used as a pre-processing step in image comprehension, eliminating redundant information and enhancing computational performance for a variety of high-level vision tasks, and it has been widely used in action recognition (Zong et al., 2021), object tracking (Hong et al., 2015), and other fields.

To the best of our knowledge, existing research on image-based SOD can be roughly categorized into classifications: RGB SOD (Fan et al., 2018a; Zhao et al., 2019), RGB-Depth (RGB-D) SOD (Liu et al., 2021; Wu et al., 2022) and RGB-Thermal (RGB-T) SOD (Cong et al., 2023; Tang et al., 2025). Compared to RGB SOD, RGB-D and RGB-T SOD task essentially introduces extra-modal that are still effective in challenge scenarios (e.g., low-light conditions, foreground-object similar to the background, and cluttered backgrounds.) to improve the generalization of the model. Unfortunately, the quality of depth images and thermal infrared images is

easily affected by their respective environments, as shown in Fig. 1. For example, depth images are difficult to provide effective information or even introduce additional noise when confronted with situations such as small objects or shallow depth of field. The thermal infrared images are easily affected by factors such as thermal crossover and thermal radiation, resulting in distortion in the image and affecting the performance of the model. Recently, considering the different applicability scenarios of depth images and thermal infrared images, researchers have developed V-D-T SOD (Cui et al., 2025; Sun et al., 2026) to enhance the applicability of the model in more scenarios, including three main research directions: multi-modal feature interaction (Huang et al., 2024; Song et al., 2023b; Wan et al., 2024a,b), images quality awareness (Bao et al., 2024) and lightweight model (Qiu et al., 2024; Song et al., 2023a). Although these V-D-T SOD methods have verified the effectiveness of the integrated use of three modals, they still suffer from the following two challenges:

1. Compared with single-modal and dual-modal SOD, V-D-T SOD inherently demands greater computational overhead due to its more

^{*} This work was supported by National Natural Science Foundation of China (NSFC) Projects (62162013, 62301453), Natural Science Foundation of Guizhou Province Science and Technology Department (QianKeHeJiChu MS-[2025]-249).

^{*} Corresponding author.

E-mail addresses: wanganzhi@gznu.edu.cn (A. Wang), cvmlwjtg@gmail.com (J. Wu), yunliu@swu.edu.cn (Y. Liu).

<https://doi.org/10.1016/j.neunet.2026.109578>

Received 12 March 2026; Received in revised form 19 July 2026; Accepted 29 August 2026

Available online 31 August 2026

0893-6080/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

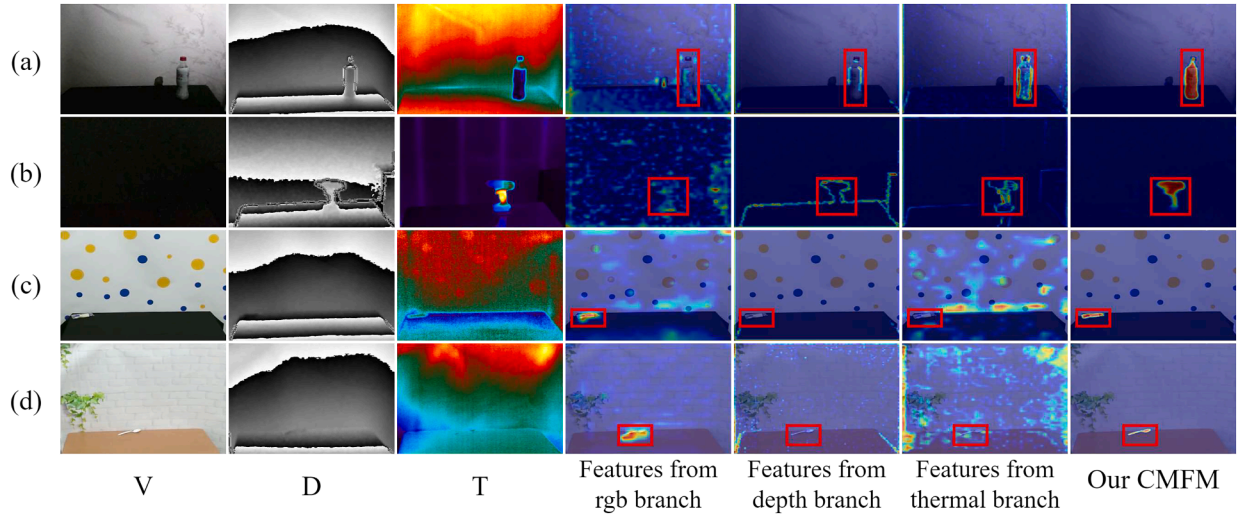


Fig. 1. Challenge examples of each modal. (a) Regular V-D-T image. (b) Low quality RGB image. (c) Low quality depth image. (d) Low quality thermal image. It can be seen that the features extracted by the network for each modal are severely affected by the quality of the input image. Simultaneously using three modal can alleviate the situation where one or two modal fail. And our CMFM can effectively extract and aggregate significant clues from three modal features.

inputs. However, existing high-performance methods usually rely on intricate network structures, which severely limit their applications on edge devices. How to design a lightweight V-D-T SOD architecture that balances efficiency and accuracy, while enhancing real-time capability and usability, remains a significant problem to be addressed.

2. Although the aforementioned methods have proven that leveraging different modalities can effectively boost performance of SOD, blindly fusing triple-modal features easily introduce noise information, leading to the degradation of algorithm performance. How to extract valid information from triple-modal features and exploit the potential of multi-modal collaboration remains an extremely challenging problem.

To address these challenges, we propose a novel lightweight attention-aware fusion network based on state-space model for V-D-T salient object detection. Specifically, to efficiently integrate triple-modal feature, we design a lightweight fusion module named Cross Mamba Fusion Module (CMFM). It utilizes the linear complexity of SSM to calculate the regions of interest within and between shallow high-resolution triple-modal features and aggregate their complementary saliency cues. Considering the inherent forgetfulness of SSM, we designed Attention Perception Fusion Module (APFM) based on self-attention mechanism to mine complete semantic relationships. Finally, we design a simplify decoder to gradually recover and generate saliency predictions from multi-level aggregated features.

The main contributions of this paper are as follows:

- We propose a novel lightweight attention-aware fusion network (LAANet) for V-D-T SOD. Unlike existing Mamba/SSM-based SOD methods that mainly employ SSM for single-modal feature modeling, our method extends SSM to cross-modal interaction and combines it with self-attention in a scale-dependent manner.
- A novel Cross Mamba Fusion Module (CMFM) is proposed to efficiently integrate multi-modal feature. CMFM simulates the process of cross attention mechanism and can achieve perception and fusion of attention regions within and between multi-modal features with only 2.39M parameters when it deployed in all layers. To compensate the inherent forgetfulness of SSM, a novel Attention Perception Fusion Module (APFM) based on self-attention mechanism is proposed. APFM can mine complete semantic relationships and has lower computational overhead than CMFM when deployed in deep networks.
- A lightweight decoder (LDB) is proposed to efficiently reconstruct image features. LDB is consist of simple dilated convolutions and linear operations with only 0.37M parameter, significantly reducing

computational overhead while maintaining high reconstruction accuracy.

- Extensive experimental results on V-D-T datasets demonstrate that our LAANet achieves close to SOTA performance and outperforms existing lightweight multi-modal SOD methods in performance and efficiency.

2. Related work

2.1. Single-modal salient object detection

SOD was initially developed on easily accessible RGB images, i.e., single-modal SOD. Early conventional single-modal SOD methods were usually based on manual features or prior knowledge to construct the model (Cheng et al., 2015; Li et al., 2013). However, due to the limited information expressed by manual features or prior knowledge, traditional methods are difficult to obtain satisfactory results or even algorithm failure when facing complex scenes. In recent years, with the rapid development of deep learning, deep neural networks represented by CNNs and ViTs have broken through the limitations of traditional methods with their powerful feature extraction capabilities, and have achieved remarkable success in various visual tasks. In 2015, Zhao et al. (2015) first introduced deep learning into salient object detection, utilizing deep neural networks to automatically learn feature representations of salient objects in images, achieving better detection performance than traditional methods. Since then, a series of outstanding studies on deep learning-based SOD have emerged. For example, Zhang et al. (2017) proposed a multi-scale deep network structure, which improves detection accuracy by fusing feature maps of different scales to capture multi-scale information of salient objects. Liu et al. (2020) utilized attention mechanisms to weight each pixel in feature maps, enabling networks to focus on salient regions and suppress background interference. Zhao et al. (2019) extracted edge information of images to guide salient object detection, enhancing detection accuracy and demonstrating the complementarity of edge information and deep features. Tu et al. (2022) introduced the DCN model, dividing the network into two stages. The decomposition network iteratively uses cross-task and cross-layer aggregation modules to predict saliency, edge, and skeleton maps. In the composition network, edge and skeleton information is used to locate the boundaries and interiors of salient objects, respectively, which helps in filling defects and suppressing noise. Li et al. (2024) explored the potential of GCN in SOD tasks. They measured feature distances in non-Euclidean

space and constructed object context relationships as topological graphs. Then, GCN is employed to compare and model context information, enabling the model to learn metrics from features and adapt to the hidden space distribution of data, thus obtaining accurate feature relationships for precise detection.

The above methods can achieve promising results in simple scenes, but they rely on appearance information such as color, contour, and texture in RGB visible images. When faced with complex situations such as similar appearance of background and foreground, insufficient image lighting, etc., algorithms are difficult to provide satisfactory detection results or even fail.

2.2. Dual-modal salient object detection

Dual-modal SOD, represented by RGB-D SOD and RGB-T SOD, introduces depth or thermal images to enhance the limitations of RGB images in challenging scenarios like low contrast and uneven lighting. Early dual-modal SOD relied on handcrafted features or prior knowledge. In the deep learning era, diverse research directions have emerged. Some studies focus on cross-modal feature fusion. For example, [Liu et al. \(2022\)](#) employed a dual-stream Swin Transformer encoder to extract multi-modal features, and subsequently proposed spatial alignment and channel re-calibration modules to optimize intra-layer cross-modal features. Additionally, to address ambiguous boundaries, the decoder facilitates inter-layer cross-modal fusion under the guidance of edge features. [Huo et al. \(2022a\)](#) employ HRFormer to expand the feature receptive field, using global optimization and attention mechanisms to select and purify modalities at the input level. They then optimize HRFormer's output features through a designed bidirectional short-connection fusion module, thereby enhancing the detailed representation of objects at the output level. [Tang et al. \(2025\)](#) proposes a "Divide-and-Conquer" strategy-based ConTriNet with three parallel flows (RGB-specific, Thermal-specific, and modality-complementary flows) to explore modality-specific cues and cross-modal complementary information. It introduces a Modality-induced Feature Modulator (MFM) and Modality-aware Dynamic Aggregation Module (MDAM) to reduce modality discrepancies and enhance robustness against defective modalities. [Wang et al. \(2025\)](#) employ an asymmetric dual-stream architecture (SwinV2-B for RGB and Res2Net50 for depth), designing a cross-modal fusion encoder to leverage the complementary properties between RGB and depth images to enrich information for salient object recognition. [Yang et al. \(2026b\)](#) propose a Modality Adaptive Network capable of handling salient object detection with arbitrary modality combinations, dynamically adjusting fusion weights for different modalities through adaptive mechanisms to address modality imbalance caused by fixed fusion strategies in traditional methods. [Yang et al. \(2026a\)](#) focus on fusion bias issues in RGB-D salient object detection, proposing solutions to mitigate bias through optimized cross-modal fusion strategies that prevent performance degradation caused by single-modality dominance. [Liu et al. \(2021\)](#) leverage a set of parameter-sharing transformer blocks to enhance both branch and aggregated features derived from a CNN architecture, thereby achieving global semantic enhancement. [Pang et al. \(2023\)](#) combined self-attention and cross attention to create a cross-modal aggregation unit, gathering info from other features in global sequences based on semantic similarity, with added channel branches for both attentions to fully utilize inter-modal and intra-modal info from spatial and channel perspectives. As shown in [Fig. 1](#), imaging devices may produce low-quality images when acquiring depth or thermal images due to environmental factors. Some research determines the usage rate of depth or thermal images from a quality assessment perspective. [Cong et al. \(2023\)](#) introduced an illuminance scoring method for low light scenes to measure image credibility, using it as a weight to control RGB and thermal image usage. [Chen et al. \(2021\)](#) obtained a pseudo label of depth image quality by calculating the similarity between segmented depth maps and saliency ground truth maps. This pseudo label served as supervision in training, enabling the

network to perceptually learn depth image quality and acquire a representation factor to control the introduction of depth information. [Fan et al. \(2021\)](#) conducted quality evaluation at the output end, deciding the final output based on the similarity between saliency maps from depth and RGB-D inputs. If differences were large, depth info was discarded.

Although depth images and thermal images can improve the performance and applicability of SOD algorithm to some extent, dual-modal still faces inherent challenges in depth images and thermal images. Depth images are ineffective or noisy for small objects or objects with little positional difference. Thermal crossover, thermal radiation, and reflection affect infrared images. These issues harm the algorithmic performance and generalization of dual-modal SOD in diverse challenging scenarios.

2.3. Triple-modal salient object detection

2.3.1. Precision model

Considering that depth and thermal images can adapt to different scenarios, some studies have begun to explore V-D-T SOD using RGB, depth, and thermal modalities. For example, [Song et al. \(2023b\)](#) created the first three modal dataset VDT2048 and proposed the first V-D-T SOD benchmark HWSI. They used three stream encoders to extract features from each modality, dilated convolution to explore complementary info within single-modal features, and attention guided aggregation of RGB-depth and depth-thermal features. Then they weighted the two aggregated features and used linear calculations and convolutions guided by global info for triple-modal feature aggregation. [Wan et al. \(2024b\)](#) proposed MFFNet, which in the encoding process, takes RGB stream as an example. Except for the fifth layer, after each layer's feature output in the encoder, MFFNet introduces corresponding depth and thermal features. Under attention guidance, it injects key info from these two modalities into RGB features, feeding them into the next layer encoder. The other two stream encoders are similarly structured. During decoding, the network progressively decodes by combining multi-scale context and edge cues to generate predictions. [Huang et al. \(2024\)](#) designed a dynamic modal weight generator for adaptive multi-modal fusion. [Wan et al. \(2024a\)](#) used dilated convolution and attention mechanisms for inter-modal optimization and interaction, adding edge supervision to refine object boundaries. [Bao et al. \(2024\)](#) addressed potential image quality issues with three subnetworks to identify high and low quality image regions and selectively fuse effective features. These methods explore the complementarity of three modal features from the perspective of feature interaction and image quality perception. However, these methods adopt complex architectural designs to achieve precise detection. This results in expensive computational costs for the model, limiting the application of V-D-T SOD on downstream tasks or resource-constrained applications.

2.3.2. Efficient model

Despite proving the effectiveness of triple-modal approaches for SOD, most existing methods use complex structures and require extra computational overhead for processing three modalities, limiting their deployment efficiency in resource-constrained environments. To this end, [Song et al. \(2023a\)](#) adopted lightweight MobilenetV2 as the backbone, designed specific modules for aggregating RGB and thermal features at different levels, and used depth images as semantic info and supervisory signals to aid feature extraction. [Qiu et al. \(2024\)](#), considering the computational burden of multi-stream encoders, designed a single stream backbone for V-D-T SOD. They improved EfficientFormerV2 by introducing a multi-modal multi-head attention module, enabling simultaneous processing and calculation of attention among three modalities within a stage and generating corresponding features. Although these methods have initially explored the lightweight architecture, the complementarity of triple-modal features has not been fully exploited.

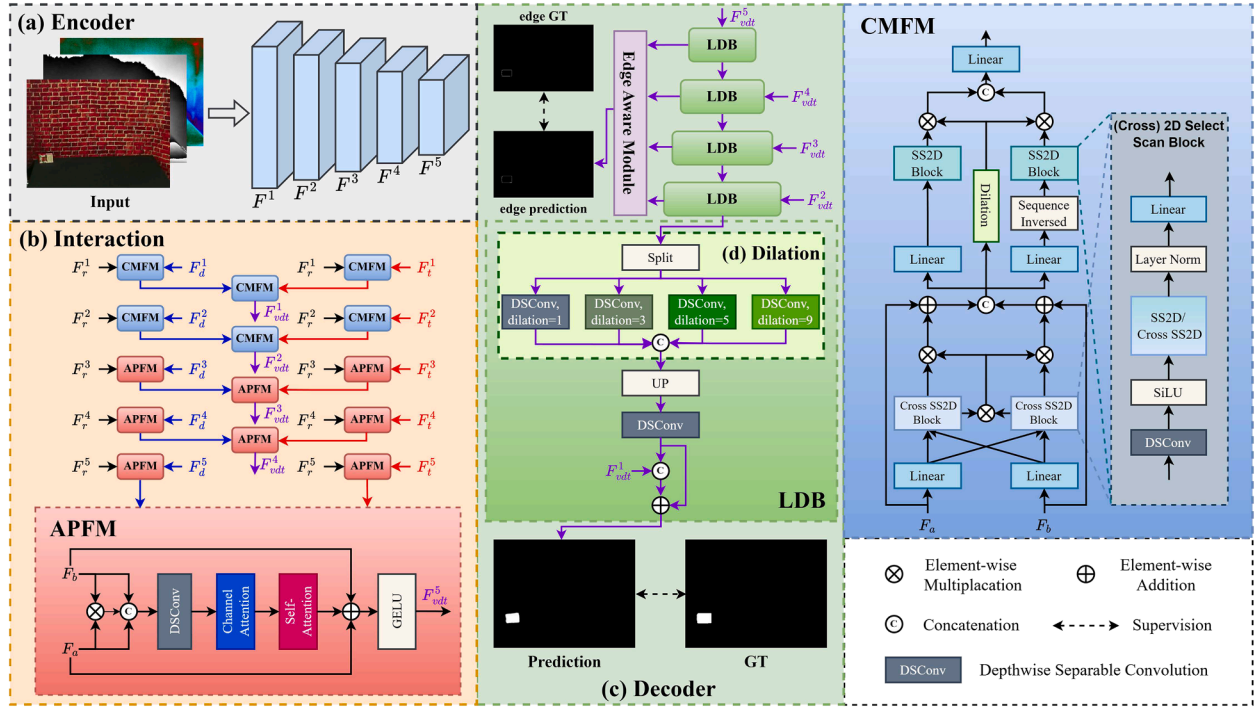


Fig. 2. The architecture of the proposed LAANet. We employ the designed fusion modules CMFM and APFM in the shallow and deep layers of the network, respectively, to efficiently integrate features from the two input modalities. We use RGB images as the primary modality, separately fuse V-D and V-T, and then aggregate the two results F_{rd}^i and F_{rt}^i using the same operation. The final result F_{rdt}^i will be fed into a lightweight decoder to progressively recover image features. At the same time, the decoded features of the last four layers, V_5 , V_4 , V_3 , and V_2 , are fed into the edge perception module to extract edge information. The decoded features generate the final saliency map under the guidance of multi-level supervision and edge supervision.

Unlike existing V-D-T SOD methods that employ a uniform fusion strategy across all stages, our LAANet adopts a scale-dependent hybrid fusion scheme according to the different characteristics of shallow and deep features. High-resolution shallow features require efficient global interaction, whereas low-resolution deep features demand more complete semantic dependency modeling. Recent SOD studies have also explored Mamba/SSM-based architectures for efficient long-range modeling. However, these methods mainly focus on single-modal feature encoding or independent intra-modal processing, and do not explicitly address cross-modal interaction in V-D-T fusion. Therefore, the novelty of our method does not lie in introducing SSM itself, but in proposing a cross-modal SSM formulation (via CMFM/Cross SS2D) and combining it with self-attention in a scale-dependent manner to provide the appropriate fusion mechanism at each stage.

3. Methodology

As shown in Fig. 2, LAANet can be roughly divided into encoding phase and decoding phase. During the encoding phase, we use the lightweight MobileNetV2-B0 (Sandler et al., 2018) as the backbone to extract multi-level features F_r^i , F_d^i and F_t^i ($i = 1, 2, 3, 4, 5$) from RGB, depth, and thermal images respectively. Since RGB images usually contain richer object cues, we take the RGB modality as the main branch and pair it with depth and thermal modalities respectively to mine complementary information, obtaining dual-modal aggregated features F_{rd}^i and F_{rt}^i . Then we combine these two dual-modal aggregated features to achieve triple-modal interaction, and the output is denoted as F_{rdt}^i . In this process, high-resolution shallow features ($i = 1, 2$) are fed into the CMFM, which uses state-space computation to capture complementary cues in dual-modal features. On the other hand, low-resolution deep features ($i = 3, 4, 5$) are processed by the APFM based on attention mechanisms to explore richer contextual semantics. In the decoding phase, the triple-modal features F_{rdt}^i are fed into a lightweight decoder com-

posed of convolutions and linear operations. The decoder progressively aggregates multi-scale features from deep to shallow layers to generate the final prediction. Meanwhile, the decoded features are passed to the EAM, which sharpens the edge details of the final prediction through auxiliary supervision.

3.1. Cross Mamba fusion module

Benefiting from the strong feature encoding capability of CNN, the backbone can extract multi-level features that contain rich object information. However, due to the limited receptive field of CNN, the features have disadvantage in global perception. To enhance the global nature of the model, we introduced the structure of the state-space model, which has recently been widely concerned. SSM can map sequence data to the state space and establish long-term feature dependencies at a linear complexity cost through its unique temporal computation. However, existing vision SSM/Mamba architectures, including recent SOD adaptations, are mainly designed for single-modal modeling or independent intra-modal processing, and therefore cannot explicitly capture cross-modal interaction in V-D-T fusion, as illustrated in Fig. 3. To this end, inspired by the cross-attention mechanism, we proposed the CMFM to enhance the perception of global dependencies while aggregating multi-modal features, and its structure is shown in Fig. 2. Our CMFM is innovatively extend SSM for cross-modal interaction through the proposed Cross Selective Scan mechanism, which enables attention-like interactions between modalities at linear complexity. This is fundamentally different from applying SSM independently to each modality. CMFM can take any backbone features of the same scale, $F_x^i \in \mathbb{R}^{B \times C \times H \times W}$ and $F_y^i \in \mathbb{R}^{B \times C \times H \times W}$, as input. In this study, it is used for shallow backbone features ($i = 1, 2$). After receiving the features, CMFM first performs a linear transformation on the two features. Following this, a DSConv layer and SiLU activation function are applied to process them. Subsequently, they are serialized into F_x^i and F_y^i , at which time the feature

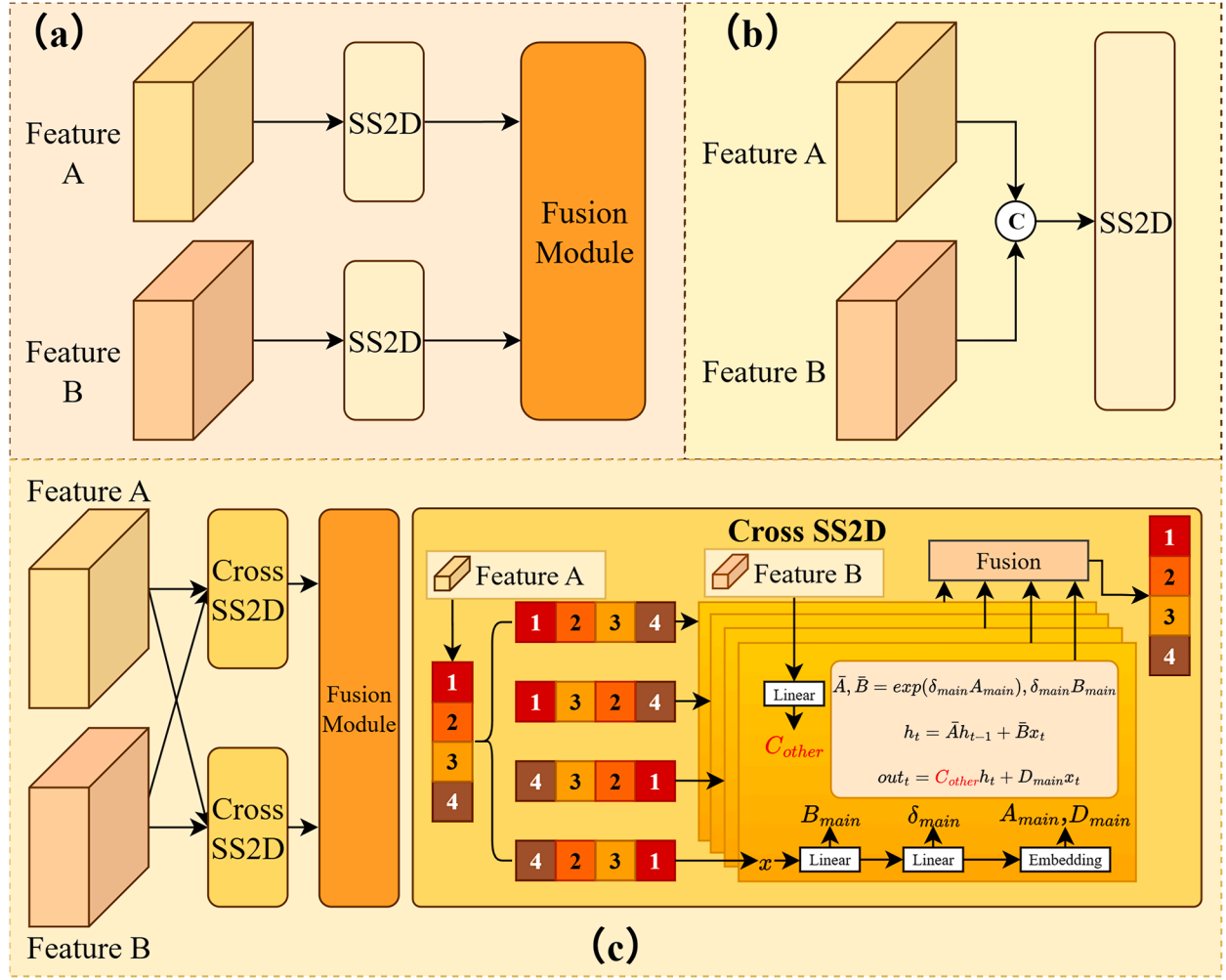


Fig. 3. Comparison between our method and existing Mamba-based SOD architectures, where (a) and (b) represent the usage strategies of SSM in current SOD research, and (c) denotes our proposed CMFM.

size is $B \times (H \times W) \times C$. The cross select scan 2D (CSS2D) block will receive the two serialized features to perform state space computation, and the computation process is shown in Algorithm 1. We construct CSS2D by swapping the observation matrix C of the four-dimensional scan module Selective Scan 2D (SS2D) to achieve spatial perception between dual-modal features and to explore their potential consistency dependencies. The above process can be defined as:

$$\begin{aligned}
 F_x^{i'} &= \text{sequence}(\text{SiLU}(\text{DSConv}(\text{Linear}(F_x^i))), \\
 F_y^{i'} &= \text{sequence}(\text{SiLU}(\text{DSConv}(\text{Linear}(F_y^i))), \\
 F_x^{i''} &= \text{Linear}(\text{LN}(\text{CSS2D}(F_x^{i'}, F_y^{i'}))), \\
 F_y^{i''} &= \text{Linear}(\text{LN}(\text{CSS2D}(F_y^{i'}, F_x^{i'}))),
 \end{aligned} \quad (1)$$

where $\text{DSConv}(\cdot)$, $\text{SiLU}(\cdot)$, $\text{sequence}(\cdot)$, $\text{LN}(\cdot)$ and $\text{CSS2D}(\cdot)$ represent the depthwise separable convolution operation, SiLU activation function, sequence operation, layer normalization, and cross select scan 2D block, respectively. Subsequently, the features from CSS2D are multiplied to mine their consistent salient cues, which enhances the original features by emphasizing important objects and suppressing background noise. By concatenating the two features, the initial aggregated feature $F_{CSS2D}^{i'}$ is obtained. This process can be formulated as:

$$\begin{aligned}
 \text{Atten} &= F_x^{i''} \otimes F_y^{i''}, \\
 F_{CSS2D}^i &= \text{Cat}((F_x^i + \text{Atten} \otimes F_x^{i''}), (F_y^i + \text{Atten} \otimes F_y^{i''})),
 \end{aligned} \quad (2)$$

where $\text{Cat}(\cdot)$ denotes the concatenation operation. To further enhance the global awareness of the aggregated features, we employ SS2D. Additionally, considering the causality of SSM, we introduce an extra reverse sequence branch for comprehensive scanning. The outputs of the two branches are concatenated, followed by a linear transformation layer and a reverse sequence operation to obtain the final output F_{CMFM}^i of CMFM. This process can be formulated as:

$$\begin{aligned}
 F_{SS2D}^i &= F_{CSS2D}^i \otimes \text{SS2D}(\text{Linear}(F_{CSS2D}^i)), \\
 F_{inSS2D}^i &= F_{CSS2D}^i \otimes \text{SS2D}(\text{Inverse}(\text{Linear}(F_{CSS2D}^i))), \\
 F_{CMFM}^i &= \text{unsequence}(\text{Linear}(\text{Cat}(F_{SS2D}^i, F_{inSS2D}^i))),
 \end{aligned} \quad (3)$$

where $\text{SS2D}(\cdot)$, $\text{Inverse}(\cdot)$ and $\text{unsequence}(\cdot)$ denote selective scan 2D operation, reverse sequence order transformation, and inverse sequencing, respectively.

3.2. Attention perception fusion module

From a causal point of view, models with Recurrent Neural Network (RNN) architectures, including SSM, memorize prior information by compressing previous tokens into fixed-size hidden states. This fixed size means that such methods are inherently loss in the way they memorize and can only access information from previous and current timesteps. In contrast, self-attentive mechanisms update causality by continuously adding the key and value of the current token, so memory is non-destructive and each token can aggregate information from all previous

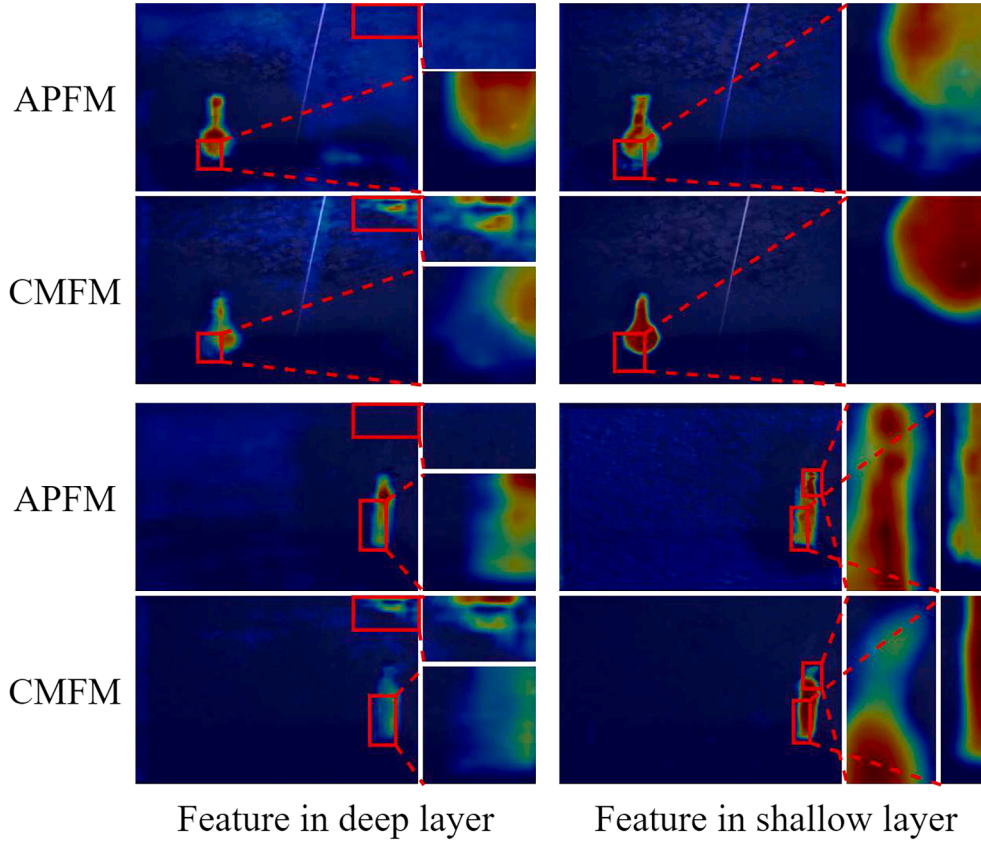


Fig. 4. Visualization of CMFM and APFM in different layer.

Algorithm 1 Cross 2D select scan module.

Input token sequence: $x^{t-1} : (B, M, D)$, $y^{t-1} : (B, M, D)$
Output token sequence: $x^t : (B, M, D)$

```

1: for  $o$  in forward, backward do
2:    $B_o^x : (B, M, N) \leftarrow \text{Linear}_B(x_o^{t-1})$ 
3:    $C_o^y : (B, M, N) \leftarrow \text{Linear}_C(y_o^{t-1})$ 
4:    $/ * C_o^y : (B, M, N) \leftarrow \text{Linear}_C(x_o^{t-1})$  when 2D SSM  $*$  /
5:    $\delta_o : (B, M, D) \leftarrow \log(1 + \exp(\text{Linear}_\delta(x_o^{t-1}))) + \text{Parameter}_\delta^o$ 
6:    $/ * \text{shape of } \text{Parameter}_\delta^o \text{ is } (D, N) * /$ 
7:    $\bar{A}_o : (B, M, D, N) \leftarrow \exp(\delta_o \otimes \text{Parameter}_\delta^o)$ 
8:    $\bar{D}_o : (B, M, N) \leftarrow \delta_o \otimes \text{Parameter}_\delta^o$ 
9:    $\bar{B}_o : (B, M, D, N) \leftarrow \delta_o \otimes B_o$ 
10:   $/ * \text{initialize } h_o \text{ and } y_o \text{ with } 0 * /$ 
11:   $h_o : (B, D, N) \leftarrow \text{zeros}(B, D, N)$ 
12:   $y_o : (B, M, D) \leftarrow \text{zeros}(B, M, D)$ 
13:   $/ * \text{SSM recurrent} * /$ 
14:  for  $i$  in  $\{0, 1, \dots, M-1\}$  do
15:     $h_o = \bar{A}_o[:, i, :, :] \odot h_o + \bar{B}_o[:, i, :, :] \odot x_o^{t-1}[:, i, :, \text{None}]$ 
16:     $y_o[:, i, :] = h_o \otimes C_o^y[:, i, :] + \bar{D}_o[:, i, :] \odot x_o^{t-1}[:, i, :]$ 
17:  end for
18: end for
19:  $x_t : (B, M, D) \leftarrow (y_{\text{forward}} + y_{\text{backward}})$ 
20: Return:  $x_t$ 

```

and subsequent tokens. For images, both prior and subsequent tokens may contain dependencies on the current token, and computing only the prior order information may lead to potential information lost, as shown in Fig. 4. Although self-attention mechanisms involve quadratic computational complexity, the low resolution deep features from the backbone network make this computationally feasible. Thus, we've developed the

APFM based on self-attention mechanisms to capture complete sequential causal relationships which is suffering losses in CMFM, as shown in Fig. 2. Specifically, the APFM first receives features $F_r^i \in \mathbb{R}^{B \times C \times H \times W}$ and $F_x^i \in \mathbb{R}^{B \times C \times H \times W}$ (where $i = 3, 4, 5$) from the backbone network. It then concatenates these two features along with their element-wise product to form an initial aggregated feature. After passing through a DSConv layer, this aggregated feature is fed into a channel attention unit to enhance the channel responses related to salient regions. Subsequently, an attention mechanism is applied to enable the aggregated feature to perceive global information, and the result is summed with the initial feature to produce the final output. The process can be formulated as:

$$F_{APFM}^i = GELU(SA(CA(DSConv(Cat(F_x^i, F_r^i, F_x^i \otimes F_r^i)))) + F_x^i + F_r^i, \quad (4)$$

where $GELU(\cdot)$, $SA(\cdot)$ and $CA(\cdot)$ denote the GELU activation function, self-attention calculation, and channel attention operation, respectively.

3.3. Lightweight decoder block

After thoroughly mining and interacting with the triple-modal multi-level features using CMFM and APFM, the aggregated features $F_{rdi=1}^i$ contain rich saliency cues. However, since these features have identical receptive fields, they limit the network's ability to handle objects of varying scales, especially small objects in the VDT2048 dataset. To address this, we design a lightweight decoder using basic linear operations (concatenation, addition) and DSConv, as shown in Fig. 2. First, the higher-level decoding features are split into four parallel branches along the channel dimension. Each branch has a DSConv layer with a different dilation rate (1,3,5,9) to enhance multi-scale object perception. The four branches are then concatenated. After an upsampling and a DSConv layer, the concatenated features are combined with the

current layer's features via skip connections to produce the final output. The process is formulated as follows:

$$\begin{aligned}
 F_{rdt}^i &= \begin{cases} F_{CMFM}^i, & i = 1, 2, \\ F_{APFM}^i, & i = 3, 4, 5, \end{cases} \\
 f_0, f_1, f_2, f_3 &= \begin{cases} Split(F_{rdt}^i), & i = 5, \\ Split(F_{rdt}^{i+1}), & i = 1, 2, 3, 4, \end{cases} \\
 f_j' &= DSCov_{d=2^j+1}(f_j), \\
 F_{dilate}^i &= DSCov(UP(Cat(f_0', f_1', f_2', f_3'))), \\
 F_{out}^i &= \begin{cases} F_{dilate}^i, & i = 5, \\ Cat(F_{rdt}^{i+1}, F_{dilate}^i) + F_{dilate}^i, & i = 1, 2, 3, 4, \end{cases}
 \end{aligned} \tag{5}$$

where $Split(\cdot)$ denotes the channel splitting operation, $DSCov_{d=2^j+1}(\cdot)$ represents depthwise separable convolution with a dilation rate of $2^j + 1$ and $UP(\cdot)$ indicates upsampling. To further sharpen the prediction boundaries and enhance the model's performance, we introduce an edge-aware module (Zhou et al., 2023a), which predicts object edge details from the decoding features F_{out}^i ($i = 2, 3, 4, 5$). The EAM helps to refine the edges of the predicted salient objects, making the final prediction more accurate and visually appealing. The formula for EAM is as follows:

$$\begin{aligned}
 P &= MLP(Maxpool(F_{out}^5)), \\
 \begin{cases} V_4 = TConv(F_{out}^4) \otimes Reshape(P), \\ V_3 = TConv(F_{out}^3) \otimes Reshape(P), \\ V_2 = TConv(F_{out}^2) \otimes Reshape(P), \end{cases} \\
 V &= DSCov(BN(V_2 + V_3 + V_4)), \\
 Edge &= DSCov(UP(V + DSCov(Drop(V)))),
 \end{aligned} \tag{6}$$

where $Maxpool(\cdot)$ and $MLP(\cdot)$ denote maximum pooling and multilayer perceptron, respectively. $TConv(\cdot)$ denotes transposed 2D convolutions. $Reshape(\cdot)$, $BN(\cdot)$ and $Drop(\cdot)$ denote respectively reshape, batch normalization and dropout operations.

3.4. Loss function

In order to improve the performance, we perform multilevel supervision and edge supervision on the network. We use hybrid loss function consists of weighted intersection over union (wIoU) loss and weighted binary cross-entropy (wBCE) loss to supervise the multilevel decoding features, which is formulated as follows.

$$L_{mask} = \sum_{i=1}^5 (L_{wBCE}(Pre(F_{out}^i), GT) + L_{wIOU}(Pre(F_{out}^i), GT)) \tag{7}$$

where $Pre(\cdot)$ contains two sets of DSConv to recover the decoded features as binary prediction maps, and GT denotes the ground truth of the image. Considering that edges are pixel-level information that does not involve the inner regions, we only supervise them using a normal binary cross-entropy loss. Our total loss is computed as follows:

$$L_{edge} = L_{BCE}(Edge, GT_{edge}) \tag{8}$$

$$L_{all} = L_{mask} + L_{edge} \tag{9}$$

where GT_{edge} denotes the ground truth of the image edge.

4. Experiment

4.1. Implementation details

To comprehensively evaluate our model, we trained and tested LAANet on the VDT2048 dataset (Song et al., 2023b), which comprises 2048 images with corresponding pixel-level annotations, including 1048 training images and 1000 testing images. All experiments were conducted on the same platform using an Intel(R) Core(TM) i9-12900K

CPU and one NVIDIA RTX 3090 GPU. Our implementation was developed in PyTorch with Python 3.10.15, CUDA 11.8, and cuDNN 8.9.2. During training, all images were resized to 384×384 , and data augmentation techniques including random flipping, cropping, and rotation were applied. The backbone of LAANet used a MobileNetV2-B0 model (Sandler et al., 2018) pre-trained on ImageNet (Deng et al., 2009), with other parameters initialized randomly. We optimized the training with the Adan optimizer (Xie et al., 2024), setting an initial learning rate of $1e-3$, decaying it to $1/20$ every 60 epochs, using betas of $[0.98, 0.92, 0.99]$, and a batch size of 4. Training took 17 hours and 53 minutes across 300 epochs. In testing, resized images were fed into the trained model to generate predictions without post-processing, then resized back to their original dimensions for evaluation. The source code and prediction results of LAANet will also be publicly available at <https://github.com/GZNU-WJT/LAANet>. For the compared methods, we used the authors' publicly released code whenever available and followed the training configurations reported in their original papers as closely as possible, including the optimizer type, learning rate, batch size, number of epochs, pretrained backbone initialization, and data augmentation strategy. All compared methods were retrained and evaluated locally on the same VDT2048 train/test split under the same software/hardware environment and metric computation pipeline.

4.2. Evaluation metrics

Following previous works (Wan et al., 2024a,b), we adopt five generally-recognized metrics for quantitative performance evaluation of the proposed method and other state-of-the-art competitors, namely Precision-Recall Curves, Mean Absolute Error (MAE), Mean F-measure ($mean F_m$), weighted F-measure ($w F_m$), Mean E-measure (E_ξ) (Fan et al., 2018b) and S-measure (S_α) (Fan et al., 2017).

The first is F-measure, which takes into account both Precision (P) and Recall (R) to evaluate the overall performance of the predicted results. Within this study, we also utilize the advanced version, weighted F-measure, which is also widely used in SOD field. The second is the Mean Absolute Error (MAE), which measures the average pixel-wise absolute difference between the predicted saliency maps and corresponding ground truth. The aforementioned two metrics have widespread applications in other pertinent areas within the domain of CV. The formulas utilized in this study align with those commonly employed in analogous research.

The third is S-measure (S_α), which measures the spatial structure similarities of predicted saliency maps compared to the corresponding ground truth from the perspective of region-aware and object-aware. Mathematically,

$$S_\alpha = \alpha * S_o + (1 - \alpha) * S_r, \tag{10}$$

where S_r denotes regional perception, S_o denotes object perception, and α is set to be 0.5 to balance the region similarity S_r and object similarity S_o as suggested in Fan et al. (2017).

We also calculate scores of mean E-measure E_ξ^{Mean} under the same protocol as the official paper (Fan et al., 2018b), which jointly captures image-level statistics and local pixel-wise correlation information to evaluate the similarity between the saliency prediction and the ground truth.

$$E_\xi^{Mean} = \frac{1}{W \times H} \sum_{i=1}^W \sum_{j=1}^H \xi_{FM}(i, j), \tag{11}$$

where ξ_{FM} stands for the enhanced alignment matrix at pixel location. W and H are width and height of the saliency map.

4.3. Comparisons with SOTA V-D-T SOD methods

To verify the validity of the model, we compared our LAANet with 15 representative SOTA SOD models, including 6 kinds of RGB-D SOD methods (i.e., MobileSal (Wu et al., 2022), MAGNet (Zhong et al.,

Table 1
Quantitative results on the VDT2048 dataset using five evaluation metrics (i.e. S_a , MAE , E_m , wF_m and $meanF_m$), $\uparrow$ and $\downarrow$ respectively indicate that larger or smaller is better. Values marked in **bold**, *italic*, underline respectively represent the best, second and third result.

Methods	Backbone	Public	VDT2048					Complexity Metrics					
			$S_m \uparrow$	$MAE \downarrow$	$E_m \uparrow$	$F_m^w \uparrow$	$mean F_m \uparrow$	Input Size	FPS $\uparrow$	Model Size(MB) $\downarrow$	Model Params(M) $\downarrow$	FLOPs(G) $\downarrow$	
Regular Methods													
V-D	TriTransNet (Liu et al., 2021)	VGG + ViT block	ACMMM'21	0.7440	0.0179	0.7643	0.5571	0.5572	256 * 256	9.72	558.9	138.76	293.99
	HiDANet (Wu et al., 2023)	ResNet50	TIP'23	0.8653	0.0058	0.9377	0.7820	0.7873	384 * 384	7.21	524.1	130.6	85.05
	ADINet (Wang et al., 2025)	SwinTransformerV2	ESWA'25	0.8792	0.0034	0.9513	0.8431	0.8006	256 * 256	14.39	543.6	109.45	25.21
	SwinNet (Liu et al., 2022)	SwinTransformer	TCSVT'22	0.8807	0.0035	0.9148	0.8481	0.7592	384 * 384	18.67	824.3	198.78	124.72
	HRTTransNet (Huo et al., 2022a)	HRFormer	TCSVT'22	0.8783	0.0031	0.9591	0.8596	0.8194	224 * 224	13.56	299.7	58.89	17.27
V-T	CAVER (Pang et al., 2023)	ResNet50	TIP'23	0.8701	0.0034	0.9584	0.8572	0.8265	384 * 384	15.44	224.2	55.79	49.19
	ConTriNet (Tang et al., 2025)	Res2Net50	TPAMI'25	0.8602	0.0035	0.9526	0.8556	0.8144	352 * 352	13.60	141.5	34.77	555.42
	HWSI (Song et al., 2023b)	VGG16	TMECH'22	0.8841	0.0030	0.9456	0.8619	0.8197	384 * 384	2.27	403.6	100.77	425.98
	QASNet (Bao et al., 2024)	SwinTransformer	TIP'24	0.8924	0.0024	0.9850	0.9057	0.8810	384 * 384	9.92	627.1	166.18	280.6
	IFENet (Bao et al., 2025)	SwinTransformer	TIP'25	0.8960	0.0025	0.9653	0.8890	0.8371	384 * 384	4.86	562.6	132.87	233.96
Our	MobilenetV2 + Mamba	-	-	0.8842	0.0029	0.9630	0.8683	0.8278	384 * 384	23.70	33.8	6.78	5.23
Lightweight Methods													
V-D	MoADNet (Jin et al., 2022)	MobilenetV2	TCSVT'22	0.7670	0.0142	0.8306	0.5734	0.5767	384 * 384	44.7	20.6	5.0	1.0
	MAGNet (Zhong et al., 2024)	MobilenetV2 + SwiftFormer	KBS'24	0.8721	0.0033	0.9588	0.8571	0.8319	384 * 384	47.22	66.8	16.07	9.91
	CSRNet (Huo et al., 2022b)	ESPNetV2	TCSVT'22	0.8529	0.0048	0.9506	0.8235	0.7934	384 * 384	41.12	4.7	1.01	6.27
V-T	LSNet (Zhou et al., 2023b)	MobilenetV2	TIP'23	0.8660	0.0040	0.9619	0.8449	0.8188	384 * 384	94.74	22.6	4.57	3.61
	MFDF (Song et al., 2023a)	MobilenetV2	JKSUCIS'23	0.8796	0.0031	0.9583	0.8644	0.8120	384 * 384	12.5	56.4	8.07	6.02
	Our	MobilenetV2 + Mamba	-	0.8842	0.0029	0.9630	0.8683	0.8278	384 * 384	23.70	33.8	6.78	5.23

2024) MoADNet (Jin et al., 2022), TriTransNet (Liu et al., 2021), HRTransNet (Huo et al., 2022a), HiDANet (Wu et al., 2023)), ADINet (Wang et al., 2025) 5 kinds of RGB-T SOD methods (i.e., LSNet (Zhou et al., 2023b), CSRNet (Huo et al., 2022b), SwinNet (Liu et al., 2022), CAVER (Pang et al., 2023), ConTriNet (Tang et al., 2025)) and 4 open-source V-D-T SOD methods (i.e., MFDFNet (Song et al., 2023a), HWSI (Song et al., 2023b), QASNet (Bao et al., 2024), IFENet (Bao et al., 2025)). Among them, MobileSal, MAGNet, MoADNet, LSNet, CSRNet and MFDFNet belong to the lightweight methods. MAGNet, TriTransNet, CAVER, HRTransNet belong to the dual-modal SOD methods with hybrid CNN-ViT architecture. For RGB-D methods, the inputs are visible and depth images, while for RGB-T methods, the inputs are visible and thermal images. To ensure a fair comparison, we retrained all compared methods locally using the authors' publicly available code and, whenever possible, the hyperparameter settings reported in the original papers. All methods were trained and evaluated on the same VDT2048 split under the same local software/hardware environment and the same metric computation pipeline. We observed that our reproduced results for several methods differed from the numbers reported in the original publications. Such discrepancies may arise from differences in software versions, hardware platforms, random seeds, preprocessing pipelines, or implementation details. Therefore, Table 1 reports the locally reproduced results for all compared methods. The input resolution used for each method follows its official implementation or original paper.

4.3.1. Quantitative comparison and discussion

To verify the effectiveness of our LAANet, Table 1 lists the comparison results between LAANet and the 13 methods on VDT2048 dataset, and the numerical results clearly demonstrate that our LAANet has achieved significant performance improvements when compared to other lightweight methods. Specifically, LAANet improves 0.52% in S_a , 6.4% in MAE, 0.49% in E_m , 0.45% in wF_m , and 1.94% in $meanF_m$ compared to the suboptimal lightweight method MFDF. Compared with both regular and lightweight dual-modal SOD approaches, our LAANet achieves superior results across all metrics. For instance, against the regular method HRTransNet (Huo et al., 2022a), LAANet improves wF_m and $meanF_m$ by 1.01% and 1.03%, respectively. When compared with regular V-D-T SOD methods, LAANet maintains performance close to theirs while dramatically enhancing efficiency. Specifically, against the SOTA method QASNet, LAANet delivers more than twice the inference speed (FPS), and exhibits substantial reductions in model size, model params, and computational complexity (FLOPs)-with FLOPs reduced by 53x. To demonstrate robustness under challenging scenarios, we provide detailed comparisons across different conditions in Tables 2, 3, and 4, which include: V-challenge (i.e., big salient object (BSO), low illumination (LI), multiple salient object (MSO), no illumination (NI), similar appearance (SA), side illumination (SI), and small salient object (SSO)), D-challenge (i.e., background interference (BI), background messy (BM), information incomplete (II), and small salient object (SSO)), and T-challenge (i.e., crossover (Cr), heat reflection (HR), and radiation dispersion (RD)). Overall, LAANet outperforms all lightweight methods and ranks just below the regular V-D-T SOD approaches HWSI and QASNet, demonstrating its potential to compete with regular SOD models. In several specific cases, LAANet even surpasses every lightweight competitor-for example, achieving the best MAE and E_m on V-BSO, and the best MAE and $meanF_m$ on V-SA. Compared with the lightweight method MFDF, LAANet improves MAE by 6.89%, 6.25%, 5.26%, and 0.09% on D-BI, D-BM, D-II, and D-SSO, respectively, placing it among the top three methods. Additionally, On T-Cr, T-HR, and T-RD, $meanF_m$ rises by 4.01%, 1.73%, and 0.45%, respectively. These results clearly evidence the effectiveness and robustness of our method across diverse challenging conditions.

To further evaluate the computational efficiency, we compare our LAANet with the state-of-the-art model in terms of frames per second (FPS), model size, number of model parameters and floating point operations (FLOPs), as shown in Table 1 column 11–16. Here, FPS stands for

Table 2
Quantitative results in V challenge. Here, $\uparrow(\downarrow)$ indicates the higher(lower) that larger(smaller) is better. The best three results in each metric are marked in **bold**, *italic*, and underline, respectively.

Methods	V-BSO			V-LI			V-MSO			V-NI			V-SA			V-SI			V-SSO		
	MAE $\downarrow$	E_m $\uparrow$	$mean F_m$ $\uparrow$	MAE $\downarrow$	E_m $\uparrow$	$mean F_m$ $\uparrow$	MAE $\downarrow$	E_m $\uparrow$	$mean F_m$ $\uparrow$	MAE $\downarrow$	E_m $\uparrow$	$mean F_m$ $\uparrow$	MAE $\downarrow$	E_m $\uparrow$	$mean F_m$ $\uparrow$	MAE $\downarrow$	E_m $\uparrow$	$mean F_m$ $\uparrow$	MAE $\downarrow$	E_m $\uparrow$	$mean F_m$ $\uparrow$
TriTransNet	0.0206	0.9423	0.8654	0.0178	0.8217	0.6019	0.0207	0.8328	0.6538	0.0279	0.8557	0.3279	0.0223	0.7721	0.5883	0.0206	0.8166	0.6090	0.0557	0.4413	0.1702
HiDANet	0.0132	0.9848	0.9188	0.0073	0.9444	0.7762	0.0088	0.9478	0.8027	0.0056	0.9869	0.7087	0.0055	0.9365	0.7841	0.0078	0.9390	0.7778	0.0016	0.7793	0.5483
MobileSal	0.0554	0.4650	0.3588	0.0121	0.8498	0.6880	0.0180	0.7623	0.6512	0.0067	0.8255	0.5917	0.0078	0.8644	0.7464	0.0113	0.8577	0.6959	0.0013	0.8391	0.5802
MAGNet	0.0068	0.9923	0.9436	0.0046	0.9696	0.8183	0.0042	0.9685	0.8342	0.0046	0.9273	0.6997	0.0026	0.9763	0.8431	0.0052	0.9624	0.8069	0.0011	0.9031	0.6793
SwinNet	0.0062	0.9934	0.9459	0.0044	0.9217	0.7553	0.0045	0.9374	0.7884	0.0054	0.8432	0.6331	0.0027	0.9348	0.7802	0.0057	0.9231	0.8069	0.0012	0.7082	0.4728
HRTTransNet	0.0063	0.9924	0.9459	0.0042	0.9614	0.8137	0.0040	0.9660	0.8318	0.0041	0.9234	0.7042	0.0025	0.9719	0.8399	0.0049	0.9652	0.8113	0.0012	0.8394	0.6037
CAVER	0.0078	0.9906	0.9372	0.0047	0.9700	0.8182	0.0049	0.9730	0.8417	0.0039	0.9392	0.7635	0.0030	0.9579	0.8137	0.0054	0.9629	0.8141	0.0010	0.9069	0.6871
CSRNet	0.0128	0.9669	0.8996	0.0062	0.9450	0.7767	0.0081	0.9501	0.8097	0.0043	0.9307	0.7338	0.0042	0.9462	0.7643	0.0077	0.9453	0.7754	0.0012	0.8415	0.5830
LSNet	0.0169	0.9690	0.8970	0.0133	0.9433	0.7873	0.0092	0.9565	0.8235	0.0051	0.8799	0.6630	0.0015	0.9654	0.8161	0.0073	0.9552	0.7990	0.0023	0.6480	0.3980
HSNet	0.0062	0.9927	0.9442	0.0048	0.9847	0.8720	0.0042	0.9738	0.8597	0.0029	0.9485	0.7879	0.0025	0.9766	0.8523	0.0039	0.9763	0.8429	0.0008	0.9136	0.7243
QASNet	0.0056	0.9931	0.9532	0.0032	0.9820	0.8617	0.0040	0.9842	0.8727	0.0031	0.9575	0.8086	0.0020	0.9896	0.8797	0.0038	0.9847	0.8645	0.0008	0.9470	0.7660
MFDF	0.0066	0.9924	0.9433	0.0041	0.9705	0.8264	0.0053	0.9636	0.8244	0.0031	0.9388	0.7636	0.0029	0.9620	0.8131	0.0045	0.9708	0.8286	0.0011	0.8702	0.6424
Our	0.0051	0.9950	0.9525	0.0049	0.9842	0.8753	0.0049	0.9710	0.8465	0.0032	0.9504	0.7670	0.0011	0.9896	0.8818	0.0040	0.9770	0.8399	0.0008	0.8908	0.6662

the number of images reasoned by the model per second, reflecting the inference speed of the model. The model size is measured in megabytes (MB) and represents the size of the memory occupied by the training model. The number of model references is measured in millions (M), indicating the storage resources consumed by the model during the inference process. FLOPs are measured in gigabytes (G), and are a quantitative measure of the computational complexity of the model. Overall, our LAANet strikes an effective balance between model efficiency and accuracy-delivering strong performance while maintaining high efficiency. Specifically, Compared to efficient dual-modal SOD methods LSNet and MAGNet, our LAANet respectively improves 2.10% and 1.39% in S_a , 2.77% and 1.31% in wF_m , while still keeping lightweight and even is more efficient than MAGNet. Moreover, our method comprehensively outperforms the lightweight V-D-T SOD approach MFDF in both performance and efficiency. Compared to the most optimal regular method QASNet, our method achieves 89.60%, 94.61%, 95.92% and 98.13% improvement in inference speed(FPS), model size, model parameters and model complexity(FLOPs), respectively. These experiments demonstrate the superiority of our method in terms of model efficiency and accuracy. In short, our method outperforms existing V-D-T SOD methods in terms of model size and, number of model parameters, and inference speed, while adding only a small amount of model complexity. Compared to the methods for dual-modal SOD, our LAANet achieves the best results in balance of efficiency and accuracy. We note, however, that these efficiency results are measured on a desktop GPU and therefore indicate deployment potential rather than direct validation on embedded edge hardware.

4.3.2. Qualitative comparison and discussion

To perform a qualitative comparison of all the saliency detection models, Figs. 5–7 present visualization results of several challenging scenarios. It can be seen that the prediction results of our model are more complete and accurate than other lightweight models, and in some cases even better than the regular method. Specifically, compared with pure CNN-based methods, our model shows a clear advantage in preserving object integrity, as shown in 5th, 9th, 11th, 13th columns in Fig. 5, 5th, 9th, 11th columns in Fig. 6 and 5th, 9th, 12th columns in Fig. 7. Due to their limited local receptive fields, CNN-based methods often fail to capture complete object shapes. This is especially true in cases of irregularly shaped objects or complex backgrounds. Our model, by effectively integrating both global and local features, can retain the shape and structure of objects more completely, such the completeness of the badminton racket detection in 1st row of Fig. 5, etc. Compared with ViT-based methods, our method excels in capturing local details as shown in 14th column in Fig. 5, and 10th column in Fig. 7. Although ViT-based methods excel at capturing global context, they may miss out on local details. Our model combines the strengths of both ViT-based global modeling and CNN-based local feature extraction, delivering superior performance in detail capture, such as the small decorations and contours of the objects in 7th row of Fig. 5, etc. Compared with existing hybrid architecture methods (MAGNet, TriTransNet, CAVER, and HRTTransNet), when dealing with multi - sized objects and edges, our model can better adapt to objects of varying sizes, generate clearer and more accurate saliency maps as shown in 3rd, 5th, 7th rows in Fig. 5, 1st, 2nd, 4th rows in Fig. 6, 1st, 3rd rows in Fig. 7. This is crucial for tasks demanding precise object localization. In summary, our model provides more comprehensive and accurate saliency detection results across a variety of challenging scenarios.

4.4. Ablation analysis

To evaluate the contribution of the main components in LAANet, we conduct ablation studies from two perspectives: component-wise evaluation and fusion-strategy analysis. As shown in Table 5, we first assess the overall impact of the proposed fusion design, LDB, and EAM by progressively adding or removing them from the baseline. In Table 6, we

Table 3

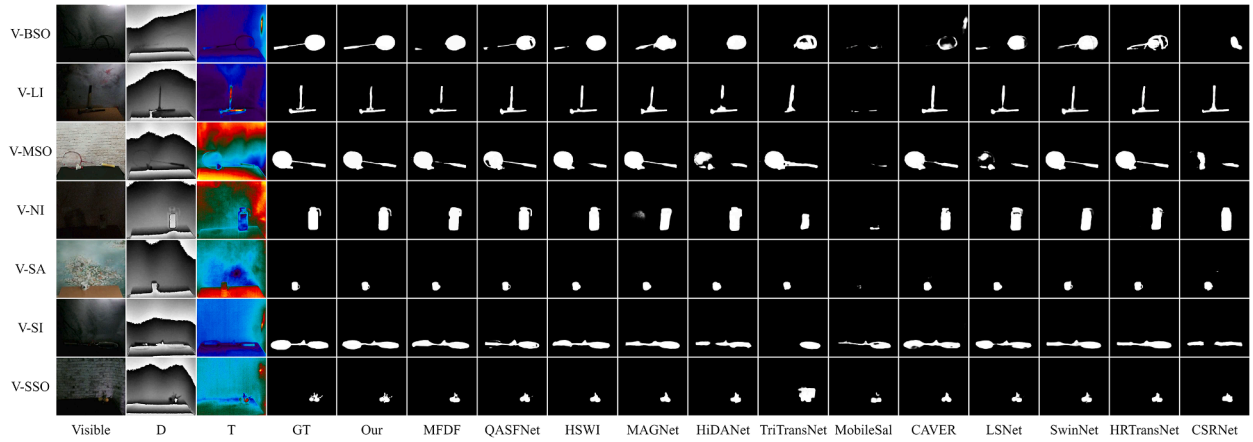
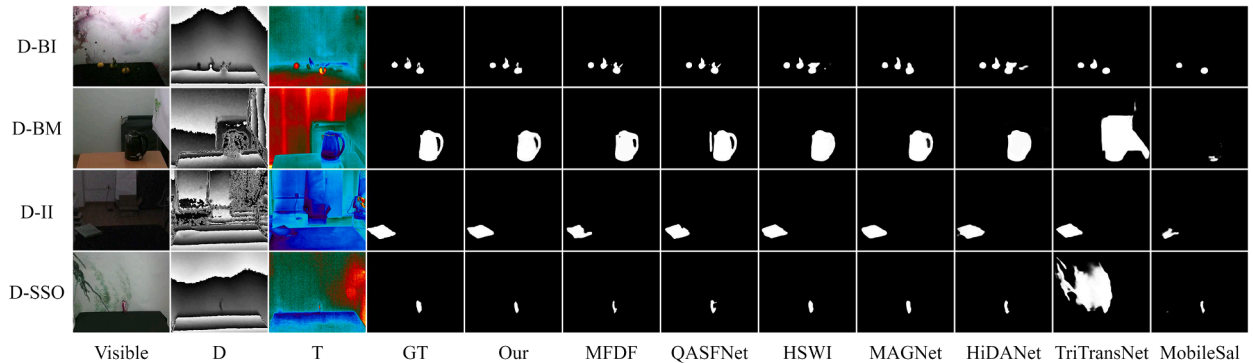
Quantitative results in D challenge. Here, $\uparrow(\downarrow)$ indicates the higher(lower) that larger(smaller) is better. The best three results in each metric are marked in **bold**, *italic*, and underline, respectively.

Methods	D-BI			D-BM			D-II			D-SSO		
	<i>MAE</i> ↓	<i>E_m</i> ↑	<i>meanF_m</i> ↑	<i>MAE</i> ↓	<i>E_m</i> ↑	<i>meanF_m</i> ↑	<i>MAE</i> ↓	<i>E_m</i> ↑	<i>meanF_m</i> ↑	<i>MAE</i> ↓	<i>E_m</i> ↑	<i>meanF_m</i> ↑
TriTransNet	0.0302	0.7425	0.5203	0.0244	0.7365	0.5149	0.0186	0.8267	0.6670	0.0557	0.4413	0.1702
HiDANet	0.0051	0.9311	0.7751	0.0063	0.9358	0.7689	0.0080	0.9580	0.8262	0.0016	0.7793	0.5483
MobileSal	0.0103	0.8345	0.6789	0.0131	0.7818	0.6314	0.0216	0.7011	0.5700	0.0013	0.8391	0.5802
MAGNet	0.0030	<u>0.9667</u>	0.8217	0.0036	0.9606	0.8086	0.0041	0.9747	0.8637	<u>0.0011</u>	0.9031	0.6793
HWSI	<i>0.0024</i>	<i>0.9778</i>	<i>0.8603</i>	<i>0.0030</i>	0.9626	<i>0.8345</i>	<i>0.0036</i>	0.9753	<i>0.8765</i>	0.0008	<i>0.9421</i>	0.7679
QASNet	0.0022	0.9849	0.8750	0.0028	0.9775	0.8633	0.0030	0.9851	0.9004	0.0008	0.9470	<i>0.7660</i>
MFDF	0.0029	0.9606	0.8163	<u>0.0032</u>	<u>0.9679</u>	0.8223	<u>0.0038</u>	<u>0.9793</u>	0.8683	<u>0.0011</u>	0.8702	0.6424
Our	<u>0.0027</u>	0.9650	<u>0.8284</u>	<u>0.0030</u>	<u>0.9678</u>	<u>0.8271</u>	<u>0.0036</u>	<u>0.9790</u>	<u>0.8757</u>	<u>0.0010</u>	<u>0.9036</u>	<u>0.6878</u>

Table 4

Quantitative results in T challenge. Here, $\uparrow(\downarrow)$ indicates the higher(lower) that larger(smaller) is better. The best three results in each metric are marked in **bold**, *italic*, and underline, respectively.

Methods	T-Cr			T-HR			T-RD		
	<i>MAE</i> ↓	<i>E_m</i> ↑	<i>meanF_m</i> ↑	<i>MAE</i> ↓	<i>E_m</i> ↑	<i>meanF_m</i> ↑	<i>MAE</i> ↓	<i>E_m</i> ↑	<i>meanF_m</i> ↑
SwinNet	0.0034	0.8932	0.7264	0.0037	0.9454	0.7834	0.0049	0.9664	0.8286
HRTransNet	<u>0.0028</u>	0.9451	0.7974	0.0036	0.9688	0.8399	0.0049	0.9805	0.8496
CAVER	0.0036	0.9522	0.7991	<u>0.0034</u>	0.9838	0.8586	0.0047	0.9859	0.8788
CSRNet	0.0049	0.9306	0.7456	0.0038	0.9742	0.8324	0.0061	0.9776	0.8522
LSNet	0.0065	0.8707	0.6670	0.0101	0.9602	0.7896	0.0062	0.9803	0.8589
HWSI	<i>0.0026</i>	0.9657	<i>0.8359</i>	<i>0.0026</i>	<i>0.9894</i>	<i>0.8876</i>	<i>0.0041</i>	<i>0.9895</i>	<i>0.8869</i>
QASNet	0.0024	0.9756	0.8520	0.0019	0.9910	0.9077	0.0035	0.9901	0.8971
MFDF	0.0034	<u>0.9676</u>	0.7990	0.0043	0.9826	0.8788	<u>0.0044</u>	0.9869	0.8754
Our	0.0033	<i>0.9699</i>	<u>0.8102</u>	0.0036	<u>0.9851</u>	<i>0.8940</i>	<u>0.0044</u>	<u>0.9870</u>	<u>0.8794</u>

**Fig. 5.** Visual comparison of V-challenge.**Fig. 6.** Visual comparison of D-challenge.

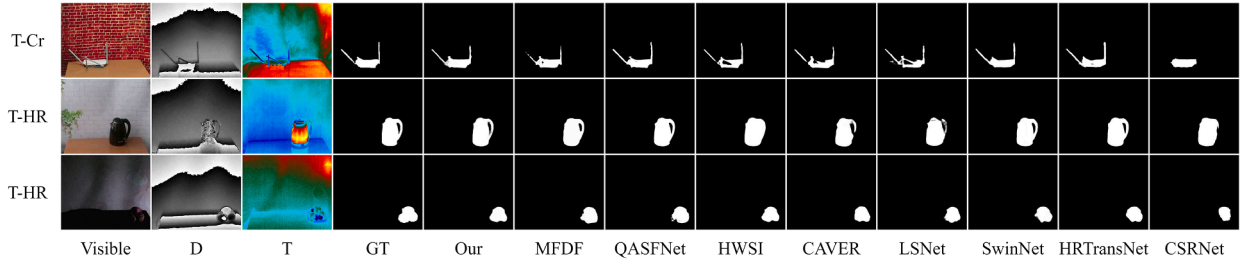


Fig. 7. Visual comparison of T-challenge.

further analyze the stage-wise deployment strategy of CMFM and APFM across network stages. Our baseline replaces the aggregation module by addition. The output is obtained from the aggregated features after sampling to 384×384 followed by splicing and convolution operations. We further visualize the results of the ablation study as shown in Fig. 8.

The Effectiveness of CMFM & APFM. First, we introduce the proposed fusion design into the baseline, as shown in Exp1 of Table 5. With the inclusion of the proposed fusion modules, all evaluation metrics are improved noticeably, while the model parameters and computational complexity increase by only 1.08M and 1.69G, respectively. We further replace the proposed fusion modules in the full LAANet with simple addition (Exp2 of Table 5). Compared with the full LAANet, this modification leads to consistent performance degradation across all metrics, with S_a and wF_m decreasing by 2.92% and 5.69%, respectively. As shown in Fig. 8, the baseline can roughly localize salient objects in some scenes, but still suffers from incomplete object structures (row 1), background noise (row 2), and even detection failure (rows 3 and 4). After introducing the proposed fusion design (Exp1), the network captures more complete object cues, such as the faint wire structure in row 1, and suppresses background interference more effectively in row 2. In addition, comparing Exp2 and Exp5 shows that replacing the proposed fusion modules with simple addition causes the predictions to exhibit issues similar to those of the baseline. This result suggests that the observed gains are not merely due to increased model capacity, but are closely related to the effectiveness of the proposed fusion mechanism itself.

We further compare several stage-wise deployment strategies of CMFM and APFM, as summarized in Table 6. The results reveal a clear trade-off between accuracy and efficiency. When APFM is applied to shallower layers, the computational cost increases significantly, and using APFM in all layers even requires reducing the input size to 192×192 due to memory limitations. In contrast, extending CMFM to deeper layers weakens semantic modeling for low-resolution features. Overall, the results indicate that neither applying CMFM to all stages nor applying APFM to all stages is optimal. Although the configuration with CMFM in layers 1, 2, and 3 and APFM in layers 4 and 5 performs better on some individual metrics, the configuration with CMFM in layers 1 and 2 and APFM in layers 3, 4, and 5 achieves a better overall balance between detection performance and efficiency, and is also more consistent with our design motivation. Therefore, we adopt the 1–2 / 3–5 configuration as the final architecture of LAANet, and all main experimental results in this paper are reported under this setting.

The Effectiveness of LDB. As with the fusion module, we show the results of adding LDBs to the baseline and removing LDBs from the full LAANet, as shown in Table 5 baseline and Exp2. It can be seen that our LDB has little computational burden on the network (only 0.6MB and 0.37M increase in the number of model size and model parameters, respectively, compared to baseline), and can effectively improve the performance of the model (E_m and wF_m increase respectively increase by 6.44% and 8.47%). Comparing Exp1 and Exp4 in Fig. 8, it can be seen that after introducing LDB, the model recognizes more object details due to its multi-scale architecture. For example, the recognition

of the wire part in row 1 is enhanced, and the object shape in row 3 is more detailed. All the above results demonstrate the effectiveness of the proposed LDB. Moreover, comparing Exp4 and Exp5 of Section 4.4, although the inclusion of EAM slightly lowers the scores on E_m and $meanF_m$, it still yields the best overall performance across all metrics, same result is displayed in Exp1 and Exp3. Further observation of the results of Exp3 compared to Exp1 and Exp5 compared to Exp4 in Fig. 8 shows that the introduction of edge information significantly improves the sharpness of object details and contours.

4.5. Failure cases and analysis

Although our LAANet can correctly detect saliency in most scenarios, it still struggles to produce satisfactory results when dealing with some challenging scenes. To give a comprehensive presentation of our LAANet, we display some failure cases in Fig. 9. For instance, in the case of multiple objects, LAANet erroneously omitted some objects or did not properly partition different objects, as shown in 1st and 2nd rows of Fig. 9. When some areas of the object are similar to the background features in the three mode image, LAANet will wrongly classify the area as the background, resulting in the loss of object integrity, as shown in 3rd rows of Fig. 9. In addition, as shown in the 4th row of Fig. 9, in cases where the boundaries are too complex, although our method can successfully locate the object body, the highlighting of object boundary details is rough and incorrect. We summarize reasons of these failures as follows:

Insufficient extraction capability of lightweight backbones. Lightweight backbones, including MobileNet adopted in this work, typically compress input images to a specified resolution before feature extraction. This process inevitably leads to substantial information loss, particularly details related to object structures, making the network unable to predict precise details.

Low-quality modality inputs. As shown in the failure case of Row 3 in Fig. 5-1, although the thermal image provides clear foreground-background discrimination, LAANet adopts an equalization strategy for multimodal information-failing to dynamically adjust weights based on modality quality. This amplifies issues such as RGB modality information failure and high similarity between target regions and background features in depth images, severely interfering with accurate object recognition and ultimately preventing the model from fully leveraging the discriminative advantages of effective modalities.

Limited training data. Despite covering some basic scenarios, the only public dataset for RGB-D-T salient object detection, VDT2048, contains merely 1048 training image pairs and 1000 testing pairs-insufficient to support model learning of generalization patterns for complex scenes. Regarding scene diversity, it only covers seven categories of home environments, lacking coverage of complex situations in real-world settings such as occlusion, abrupt illumination changes, and multi-object interactions. These limitations result in a generalization bottleneck for RGB-D-T salient object detection models, directly restricting their adaptability across various complex scenarios.

Table 5

Ablation experiment of our LAANet on the VDT2048 dataset. $\uparrow(\downarrow)$ indicates the higher (lower) that larger (small) is better. The best three results in each metric are marked in **red** and **blue**, respectively.

Component				VDT2048						Complexity Metrics(Input Size 384*384)			
CMFM & APFM	LDB	EAM		$S_a \uparrow$	$MAE \downarrow$	$E_m \uparrow$	$wF_m \uparrow$	$meanF_m \uparrow$	$maxF_m \uparrow$	$FPS \uparrow$	$ModelSize(MB) \downarrow$	$ModelParams(M) \downarrow$	$FLOPs(G) \downarrow$
baseline				0.8521	0.0044	0.8991	0.7956	0.7485	0.8805	46.84	32.4	5.47	9.06
Exp1	✓			0.8712	0.0032	0.9384	0.8445	0.7920	0.8983	26.09	31.6	6.41	10.69
Exp2		✓	✓	0.8584	0.0039	0.9570	0.8197	0.8021	0.8694	45.04	33.8	5.84	3.7
Exp3	✓		✓	0.8833	0.0030	0.9305	0.8630	0.7864	0.9044	14.58	32.4	6.46	17.07
Exp4	✓		✓	0.8743	0.0031	0.9778	0.8617	0.8521	0.8992	20.84	33.1	6.74	5.05
Exp5	✓	✓	✓	0.8842	0.0029	0.9630	0.8683	0.8278	0.8996	23.70	33.9	6.78	5.23

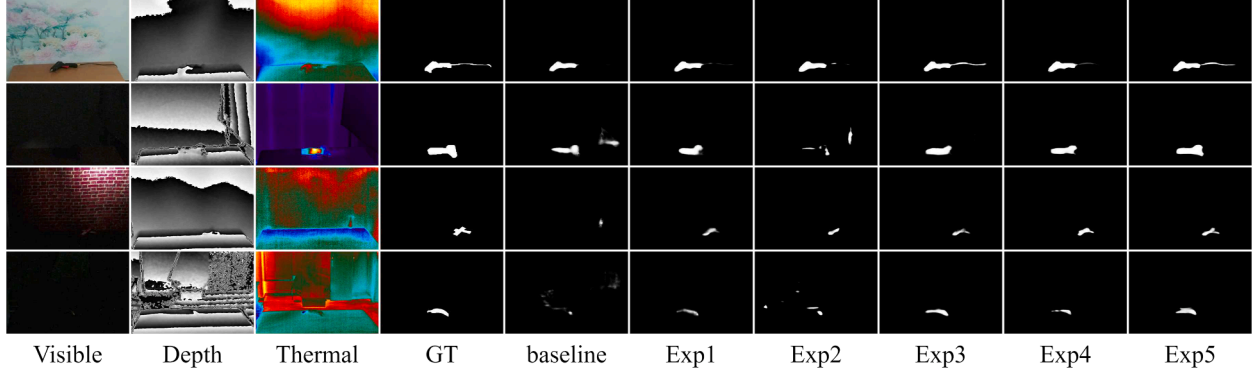


Fig. 8. Visualization of ablation studies for important components, "Exp x" corresponds to the experiment in Table 5.

Table 6

Ablation experiment of fusion strategy. $\uparrow(\downarrow)$ indicates the higher(lower) that larger(smaller) is better. Values marked in red represent the best results and blue denotes the second. Input Size smaller than 384*384 indicates that the input needs to be adjusted to the corresponding size in order to successfully train and infer.

	CMFM	APFM	Input Size	VDT2048						Complexity Metrics			
				$S_a \uparrow$	$MAE \downarrow$	$E_m \uparrow$	$wF_m \uparrow$	$meanF_m \uparrow$	$maxF_m \uparrow$	$FPS \uparrow$	$ModelSize(MB) \downarrow$	$ModelParams(M) \downarrow$	$FLOPs(G) \downarrow$
Layer	1,2,3,4,5	-	384 * 384	0.8688	0.0035	0.9580	0.8461	0.8198	0.8923	13.41	39.7	7.86	6.24
	1,2,3,4	5	384 * 384	0.8753	0.0032	0.9566	0.8560	0.8224	0.8951	15.24	34.5	6.91	5.73
	1,2,3	4,5	384 * 384	0.8804	0.0029	0.9685	0.8682	0.8399	0.9026	17.07	33.9	6.79	5.51
	1,2	3,4,5	384 * 384	0.8842	0.0029	0.9630	0.8683	0.8278	0.8996	23.70	33.8	6.78	5.23
	1	2,3,4,5	352 * 352	0.8327	0.0053	0.8983	0.7605	0.7341	0.8438	12.07	33.8	6.78	4.30
	-	1,2,3,4,5	192 * 192	0.8375	0.0061	0.8177	0.7239	0.6262	0.8333	9.50	33.7	6.78	1.13

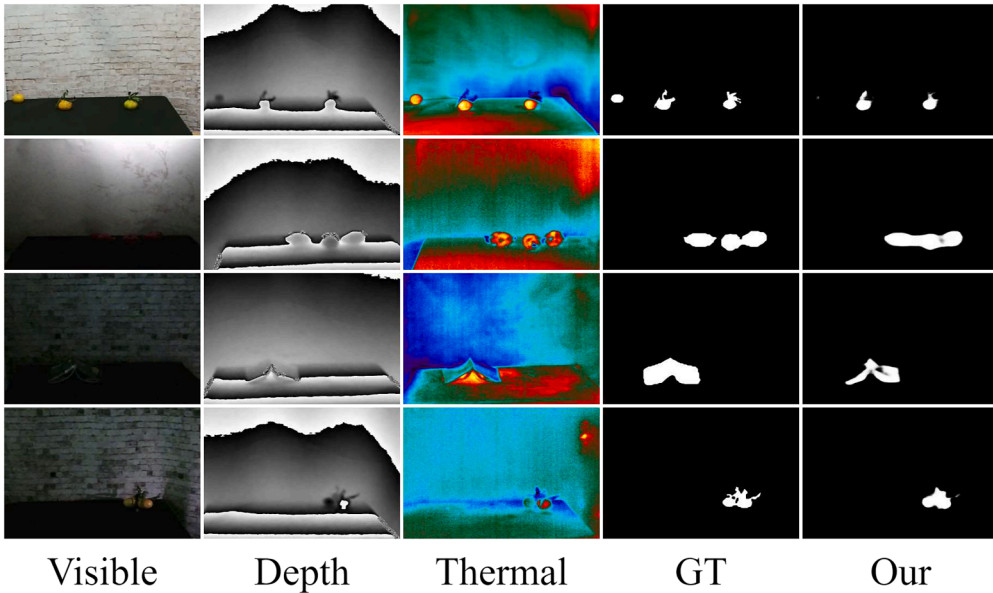


Fig. 9. Example of failure cases, including: multi objects (1st-2nd row), background interference (3rd row) and complex contours (4th row).

5. Conclusion

In this paper, we proposed a novel lightweight attention-aware fusion network (LAANet) for V-D-T SOD. In the proposed cross-modal Cross SS2D formulation in CMFM and the scale-dependent hybrid use of SSM and self-attention. We design CMFM based on the SSM architecture to efficiently aggregate shallow, high-resolution features by mimicking cross-attention mechanisms. The proposed APFM compensates for the memory loss inherent in SSM by processing the low-resolution features of the deep network through a self-attention mechanism. To guide accurate predictions, we build a lightweight decoder using DSConv and linear operations. Overall, compared with other 15 state-of-the-art SOTA methods, our LAANet demonstrates its superiority and promising potential for deployment in resource-constrained scenarios in terms of both model performance and efficiency. Based on the analysis of the LAANet failure case described above, we will consider three major development directions: 1) Constructing a ten-thousand-scale RGB-D-T dataset covering complex environments such as outdoor and industrial settings. 2) Establishing a quality evaluation mechanism based on metrics like clarity and signal-to-noise ratio to dynamically adjust fusion weights. 3) Addressing the issue of detail loss in lightweight backbone networks by designing a dynamic resolution adaptation mechanism based on parallel computing or neural architecture search.

CRedit authorship contribution statement

Anzhi Wang: Writing – review & editing, Resources, Project administration, Formal analysis, Conceptualization; **Jintao Wu:** Writing – original draft, Visualization, Validation; **Yun Liu:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- Bao, L., Zhou, X., Lu, X., Sun, Y., Yin, H., Hu, Z., Zhang, J., & Yan, C. (2024). Quality-aware selective fusion network for v-d-t salient object detection. *IEEE Transactions on Image Processing*, 33, 3212–3226. <https://doi.org/10.1109/TIP.2024.3393365>
- Bao, L., Zhou, X., Zheng, B., Cong, R., Yin, H., Zhang, J., & Yan, C. (2025). Ifenet: Interaction, fusion, and enhancement network for v-d-t salient object detection. *IEEE Transactions on Image Processing*, 34, 483–494. <https://doi.org/10.1109/TIP.2025.3527372>
- Chen, Z., Cong, R., Xu, Q., & Huang, Q. (2021). Dpanet: Depth potentiality-aware gated attention network for rgb-d salient object detection. *IEEE Transactions on Image Processing*, 30, 7012–7024. <https://doi.org/10.1109/TIP.2020.3028289>
- Cheng, M.-M., Mitra, N. J., Huang, X., Torr, P. H. S., & Hu, S.-M. (2015). Global contrast based salient region detection. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 37(3), 569–582. <https://doi.org/10.1109/TPAMI.2014.2345401>
- Cong, R., Zhang, K., Zhang, C., Zheng, F., Zhao, Y., Huang, Q., & Kwong, S. (2023). Does thermal really always matter for RGB-t salient object detection? *IEEE Transactions on Multimedia*, 25, 6971–6982. <https://doi.org/10.1109/TMM.2022.3216476>
- Cui, F., Zhou, X., Bao, L., Wan, B., Shi, R., Chen, Q., & Zhang, J. (2025). Multi-modal feature integration network for visible-depth-thermal salient object detection. *Engineering Applications of Artificial Intelligence*, 152, 110734. <https://doi.org/10.1016/j.engappai.2025.110734>
- Deng, J., Dong, W., Socher, R., Li, L.-J., Li, K., & Fei-Fei, L. (2009). Imagenet: A large-scale hierarchical image database. In *2009 IEEE Conference on computer vision and pattern recognition* (pp. 248–255). <https://doi.org/10.1109/CVPR.2009.5206848>
- Fan, D.-P., Cheng, M.-M., Liu, J.-J., Gao, S.-H., Hou, Q., & Borji, A. (2018a). Salient objects in clutter: Bringing salient object detection to the foreground. In V. Ferrari, M. Hebert, C. Sminchisescu, & Y. Weiss (Eds.), *Computer vision – ECCV 2018* (pp. 196–212). Cham: Springer International Publishing. https://doi.org/10.1007/978-3-030-01267-0_12
- Fan, D.-P., Cheng, M.-M., Liu, Y., Li, T., & Borji, A. (2017). Structure-measure: A new way to evaluate foreground maps. In *2017 IEEE International conference on computer vision (ICCV)* (pp. 4558–4567). <https://doi.org/10.1109/ICCV.2017.487>
- Fan, D.-P., Gong, C., Cao, Y., Ren, B., Cheng, M.-M., & Borji, A. (2018b). Enhanced-alignment measure for binary foreground map evaluation. In *Proceedings of the twenty-seventh international joint conference on artificial intelligence, IJCAI-18* (pp. 698–704). International Joint Conferences on Artificial Intelligence Organization. <https://doi.org/10.24963/ijcai.2018/97>
- Fan, D.-P., Lin, Z., Zhang, Z., Zhu, M., & Cheng, M.-M. (2021). Rethinking RGB-d salient object detection: Models, data sets, and large-scale benchmarks. *IEEE Transactions on Neural Networks and Learning Systems*, 32(5), 2075–2089. <https://doi.org/10.1109/TNNLS.2020.2996406>
- Hong, S., You, T., Kwak, S., & Han, B. (2015). Online tracking by learning discriminative saliency map with convolutional neural network. In *Proceedings of the 32nd international conference on machine learning - volume 37 ICML'15* (p. 597–606). JMLR.org.
- Huang, N., Yang, Y., Xi, R., Zhang, Q., Han, J., & Huang, J. (2024). Salient object detection from arbitrary modalities. *IEEE Transactions on Image Processing*, 33, 6268–6282. <https://doi.org/10.1109/TIP.2024.3486225>
- Huo, F., Zhu, X., Zhang, L., Liu, Q., & Shu, Y. (2022a). Efficient context-guided stacked refinement network for RGB-t salient object detection. *IEEE Transactions on Circuits and Systems for Video Technology*, 32(5), 3111–3124. <https://doi.org/10.1109/TCSVT.2021.3102268>
- Huo, F., Zhu, X., Zhang, L., Liu, Q., & Shu, Y. (2022b). Efficient context-guided stacked refinement network for RGB-t salient object detection. *IEEE Transactions on Circuits and Systems for Video Technology*, 32(5), 3111–3124. <https://doi.org/10.1109/TCSVT.2021.3102268>
- Jin, X., Yi, K., & Xu, J. (2022). MoADNet: Mobile asymmetric dual-stream networks for real-time and lightweight RGB-d salient object detection. *IEEE Transactions on Circuits and Systems for Video Technology*, 32(11), 7632–7645. <https://doi.org/10.1109/TCSVT.2022.3180274>
- Li, C., Liu, F., Tian, Z., Du, S., & Wu, Y. (2024). Dagcn: Dynamic and adaptive graph convolutional network for salient object detection. *IEEE Transactions on Neural Networks and Learning Systems*, 35(6), 7612–7626. <https://doi.org/10.1109/TNNLS.2022.3219245>
- Li, X., Lu, H., Zhang, L., Ruan, X., & Yang, M.-H. (2013). Saliency detection via dense and sparse reconstruction. In *2013 IEEE International conference on computer vision* (pp. 2976–2983). <https://doi.org/10.1109/ICCV.2013.370>
- Liu, N., Han, J., & Yang, M.-H. (2020). PiCANet: Pixel-wise contextual attention learning for accurate saliency detection. *IEEE Transactions on Image Processing*, 29, 6438–6451. <https://doi.org/10.1109/TIP.2020.2988568>
- Liu, Z., Tan, Y., He, Q., & Xiao, Y. (2022). Swinnet: Swin transformer drives edge-aware RGB-d and RGB-t salient object detection. *IEEE Transactions on Circuits and Systems for Video Technology*, 32(7), 4486–4497. <https://doi.org/10.1109/TCSVT.2021.3127149>
- Liu, Z., Wang, Y., Tu, Z., Xiao, Y., & Tang, B. (2021). Tritransnet: RGB-d salient object detection with a triplet transformer embedding network. In *Proceedings of the 29th ACM international conference on multimedia (ACMMM MM '21* (p. 4481–4490). New York, NY, USA: Association for Computing Machinery. <https://doi.org/10.1145/3474085.3475601>
- Pang, Y., Zhao, X., Zhang, L., & Lu, H. (2023). Caver: Cross-modal view-mixed transformer for bi-modal salient object detection. *IEEE Transactions on Image Processing*, 32, 892–904. <https://doi.org/10.1109/TIP.2023.3234702>
- Qiu, J., Jiang, C., & Wang, H. (2024). Etfomer: An efficient transformer based on multi-modal hybrid fusion and representation learning for rgb-d-t salient object detection. *IEEE Signal Processing Letters*, 31, 2930–2934. <https://doi.org/10.1109/LSP.2024.3465351>
- Sandler, M., Howard, A., Zhu, M., Zhmoginov, A., & Chen, L.-C. (2018). Mobilenetv2: Inverted residuals and linear bottlenecks. In *2018 IEEE/CVF Conference on computer vision and pattern recognition* (pp. 4510–4520). <https://doi.org/10.1109/CVPR.2018.00474>
- Song, K., Wang, H., Zhao, Y., Huang, L., Dong, H., & Yan, Y. (2023a). Lightweight multi-level feature difference fusion network for RGB-d-t salient object detection. *Journal of King Saud University - Computer and Information Sciences*, 35(8), 101702. <https://doi.org/10.1016/j.jksuci.2023.101702>
- Song, K., Wang, J., Bao, Y., Huang, L., & Yan, Y. (2023b). A novel visible-depth-thermal image dataset of salient object detection for robotic visual perception. *IEEE/ASME Transactions on Mechatronics*, 28(3), 1558–1569. <https://doi.org/10.1109/TMECH.2022.3215909>
- Sun, Y., Wan, B., Yin, H., & Chen, Y. (2026). DMDNet: Dual-branch multi-modal deep fusion network for v-d-t salient object detection. *Neural Networks*, 198, 108579. <https://doi.org/10.1016/j.neunet.2026.108579>
- Tang, H., Li, Z., Zhang, D., He, S., & Tang, J. (2025). Divide-and-conquer: Confluent triple-flow network for RGB-t salient object detection. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 47(3), 1958–1974. <https://doi.org/10.1109/TPAMI.2024.3511621>
- Tu, Z., Li, Z., Li, C., & Tang, J. (2022). Weakly alignment-free RGBT salient object detection with deep correlation network. *IEEE Transactions on Image Processing*, 31, 3752–3764. <https://doi.org/10.1109/TIP.2022.3176540>
- Wan, B., Lv, C., Zhou, X., Sun, Y., Zhu, Z., Wang, H., & Yan, C. (2024a). Tmnet: Triple-modal interaction encoder and multi-scale fusion decoder network for v-d-t salient object detection. *Pattern Recognition*, 147, 110074. <https://doi.org/10.1016/j.patcog.2023.110074>
- Wan, B., Zhou, X., Sun, Y., Wang, T., Lv, C., Wang, S., Yin, H., & Yan, C. (2024b). Mffnet: Multi-modal feature fusion network for v-d-t salient object detection. *IEEE Transactions on Multimedia*, 26, 2069–2081. <https://doi.org/10.1109/TMM.2023.3291823>
- Wang, F., Li, Y., Wang, L., & Zheng, P. (2025). Asymmetric deep interaction network for RGB-d salient object detection. *Expert Systems with Applications*, 266, 126083. <https://doi.org/10.1016/j.eswa.2024.126083>
- Wu, Y.-H., Liu, Y., Xu, J., Bian, J.-W., Gu, Y.-C., & Cheng, M.-M. (2022). Mobilesal: Extremely efficient RGB-d salient object detection. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(12), 10261–10269. <https://doi.org/10.1109/TPAMI.2021.3134684>
- Wu, Z., Allibert, G., Meriaudeau, F., Ma, C., & Demoncaux, C. (2023). HiDAnet: RGB-d salient object detection via hierarchical depth awareness. *IEEE Transactions on Image Processing*, 32, 2160–2173. <https://doi.org/10.1109/TIP.2023.3263111>

- Xie, X., Zhou, P., Li, H., Lin, Z., & Yan, S. (2024). Adan: Adaptive nesterov momentum algorithm for faster optimizing deep models. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(12), 9508–9520. <https://doi.org/10.1109/TPAMI.2024.3423382>
- Yang, Y., Huang, N., Zhang, Q., & Han, J. (2026a). Mitigating fusion bias for RGB-d salient object detection. *Pattern Recognition*, 171, 112135. <https://doi.org/10.1016/j.patcog.2025.112135>
- Yang, Y., Huang, N., Zhang, Q., Han, J., & Huang, J. (2026b). Modality adaptive network for arbitrary modality salient object detection. *IEEE Transactions on Multimedia*, (pp. 1–14). <https://doi.org/10.1109/TMM.2026.3651119>
- Zhang, P., Wang, D., Lu, H., Wang, H., & Ruan, X. (2017). Amulet: Aggregating multi-level convolutional features for salient object detection. In *2017 IEEE International conference on computer vision (ICCV)* (pp. 202–211). <https://doi.org/10.1109/ICCV.2017.31>
- Zhao, J., Liu, J.-J., Fan, D.-P., Cao, Y., Yang, J., & Cheng, M.-M. (2019). Egnet: Edge guidance network for salient object detection. In *2019 IEEE/CVF International conference on computer vision (ICCV)* (pp. 8778–8787). <https://doi.org/10.1109/ICCV.2019.00887>
- Zhao, R., Ouyang, W., Li, H., & Wang, X. (2015). Saliency detection by multi-context deep learning. In *2015 IEEE Conference on computer vision and pattern recognition (CVPR)* (pp. 1265–1274). <https://doi.org/10.1109/CVPR.2015.7298731>
- Zhong, M., Sun, J., Ren, P., Wang, F., & Sun, F. (2024). Magnet: Multi-scale awareness and global fusion network for rgb-d salient object detection. *Knowledge-Based Systems*, 299, 112126. <https://doi.org/10.1016/j.knosys.2024.112126>
- Zhou, W., Sun, F., Jiang, Q., Cong, R., & Hwang, J.-N. (2023a). Wavenet: Wavelet network with knowledge distillation for RGB-t salient object detection. *IEEE Transactions on Image Processing*, 32, 3027–3039. <https://doi.org/10.1109/TIP.2023.3275538>
- Zhou, W., Zhu, Y., Lei, J., Yang, R., & Yu, L. (2023b). Lsnet: Lightweight spatial boosting network for detecting salient objects in rgb-thermal images. *IEEE Transactions on Image Processing*, 32, 1329–1340. <https://doi.org/10.1109/TIP.2023.3242775>
- Zong, M., Wang, R., Chen, X., Chen, Z., & Gong, Y. (2021). Motion saliency based multi-stream multiplier resnets for action recognition. *Image and Vision Computing*, 107, 104108. <https://doi.org/10.1016/j.imavis.2021.104108>